

**10 (a)** The Sun has a surface temperature of 5780 K. The luminosity of the Sun is  $3.85 \times 10^{26} \text{ W}$ .

(i) Calculate the radius of the Sun.

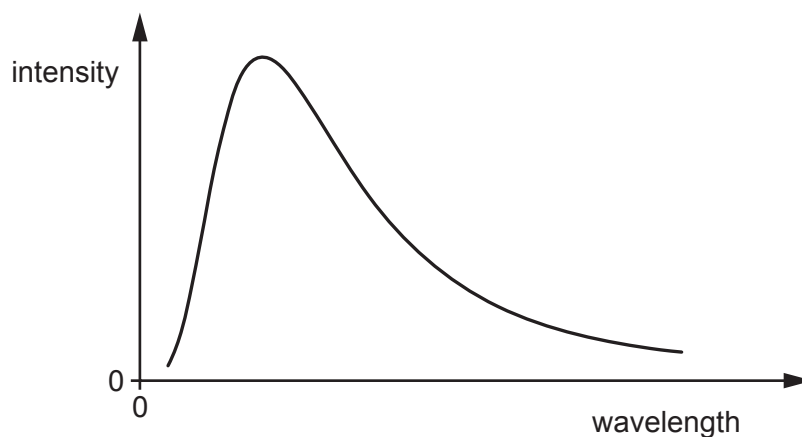
radius = ..... m [2]

(ii) The Earth is a distance of  $1.50 \times 10^{11} \text{ m}$  from the Sun.

Calculate the radiant flux intensity  $F$  of the radiation from the Sun at a distance of  $1.50 \times 10^{11} \text{ m}$ . Give a unit with your answer.

$F = \dots\dots\dots$  unit ..... [2]

(iii) The variation with wavelength of the intensity of radiation emitted from the Sun is shown in Fig. 10.1.



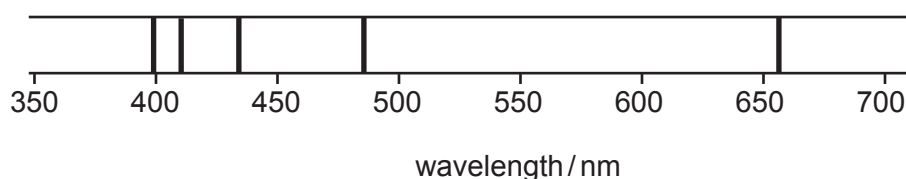
**Fig. 10.1**

Another star has the same radius as the Sun but has a lower surface temperature.

On Fig. 10.1, sketch a line to show the variation with wavelength of the intensity of the radiation emitted for this star. [2]

(b) A galaxy in the constellation Corona Borealis is moving away from the Earth.

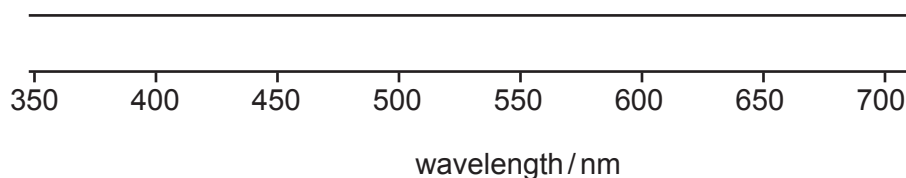
(i) The visible emission spectrum for the Sun is shown in Fig. 10.2.



**Fig. 10.2**

The lines are at wavelengths of 397 nm, 410 nm, 434 nm, 486 nm and 656 nm. The compositions of the Sun and a star in the Corona Borealis galaxy are similar.

On Fig. 10.3, sketch the emission spectrum for the star in the Corona Borealis galaxy as observed from the Earth. No calculations are required.



**Fig. 10.3**

[1]

(ii) The galaxy in Corona Borealis is moving away from the Earth at a speed of  $21\,400\text{ km s}^{-1}$ .

Use information from (b)(i) to calculate, in nm, the observed wavelength of the lowest visible energy emission for the star in the Corona Borealis galaxy.

wavelength = ..... nm [2]

(iii) The wavelength in (b)(ii) is used to calculate a value for the surface temperature of the star in the Corona Borealis galaxy. The calculation does not give an accurate value.

State and explain whether this value of temperature is too high or too low.

.....  
 .....  
 ..... [1]

[Total: 10]